



Ukraine towards Carbon Neutrality

**WP4: Pilot Implementation in Ukraine,
supported by European front-runners**

U_CAN Consortium Meeting

31.03.2026, online

Pavlo Radiuk (Khmelnytskyi National
University)



Funded by
the European Union



Manuscript Acceptance and Methodological Framing

Achievement Unlocked: Manuscript Accepted

Official acceptance to a high-impact factor (Q2) academic peer-reviewed journal.
Status: Soon available online.

March 16–17: Revised manuscript introduction, research gap, and methodological framing for topography-aware control.

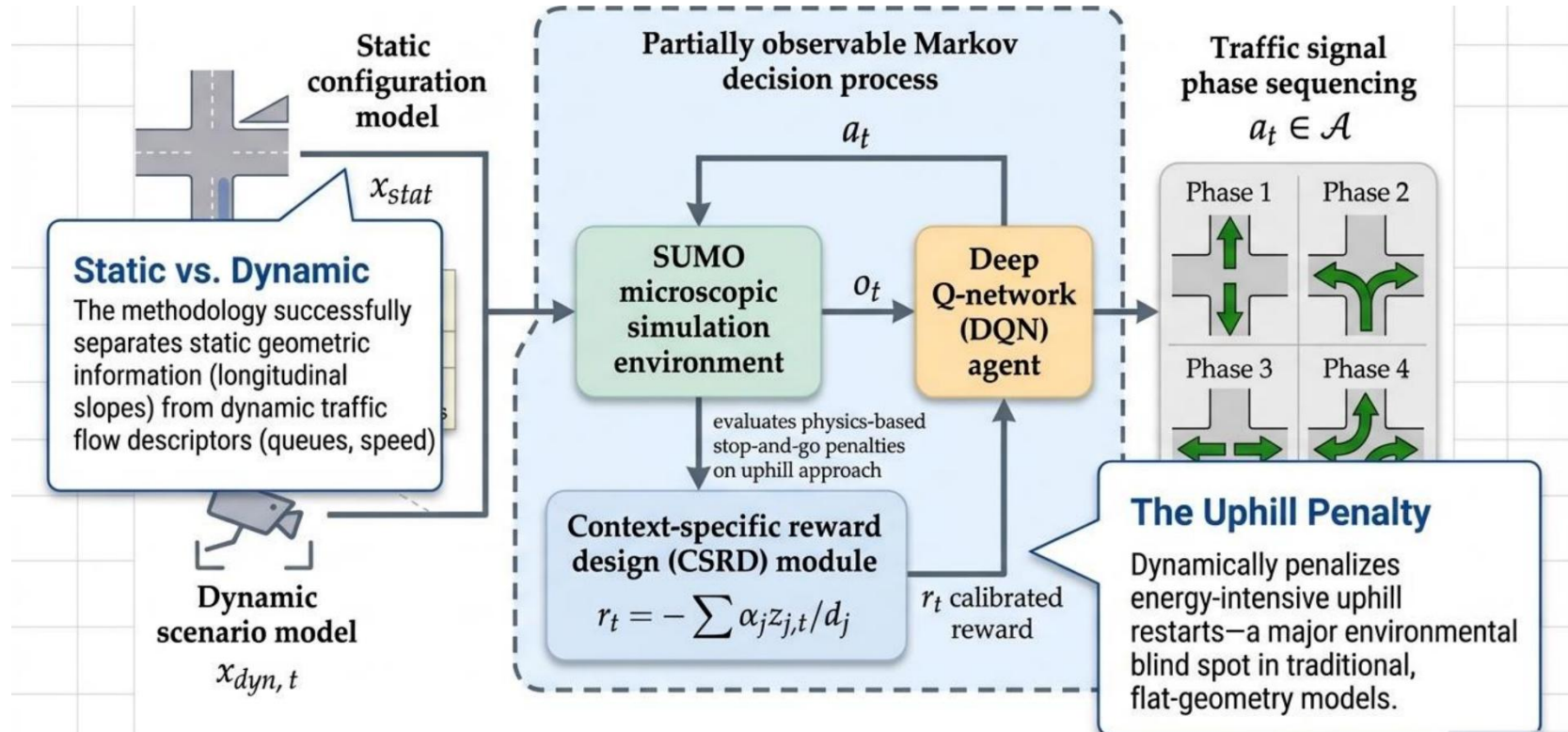


The Core Problem: Generic delay-focused reinforcement learning rewards misalign with the genuine energetic realities of hilly urban corridors.



The Solution: Topography-aware urban traffic control explicitly targeted at carbon dioxide emissions reduction.

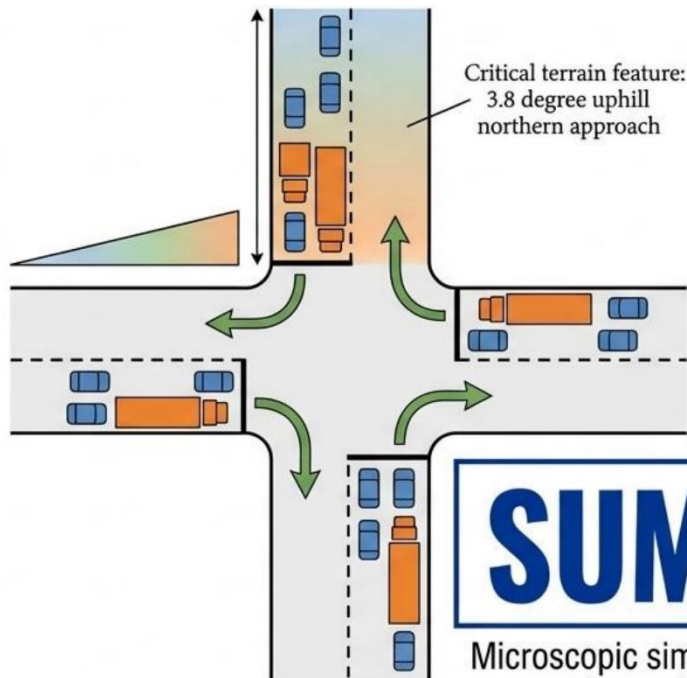
Formal Model and Reward Design Updates



Simulation Framework and Calibration Workflow

14,800

Vehicles simulated over a
9-hour horizon.



20%

Heavy-vehicle share integrated
using specific HBEFA profiles on a
challenging 3.8° uphill approach.

SUMO + DRL

Microscopic simulator coupled with a continuous
deep reinforcement learning loop.

Finalisation, Statistical Validation, and Dissemination

256.97M mg

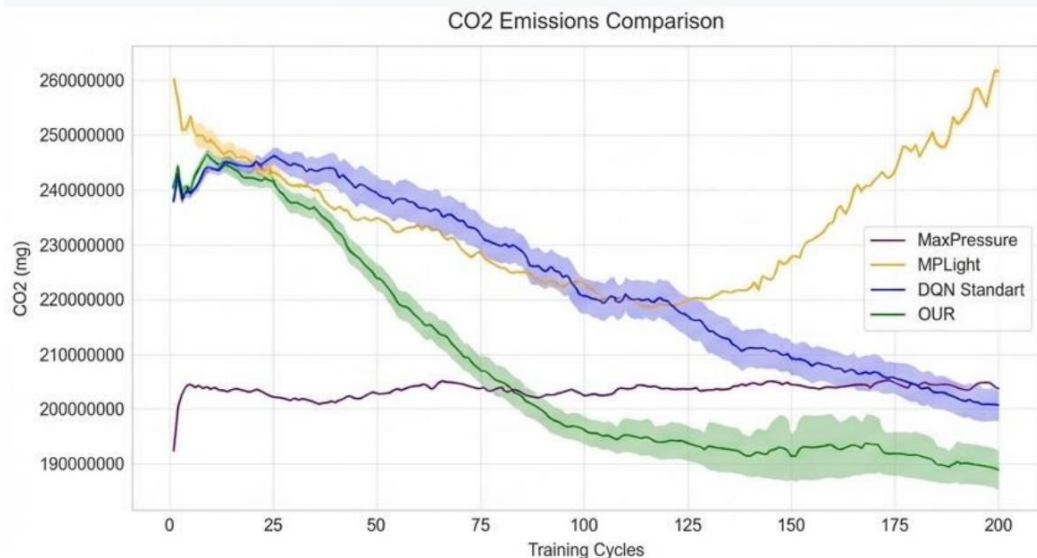
Total CO2 Emissions.

8.6%

Emissions reduction vs.
MaxPressure baseline (4.7%
vs. Standard DQN)

72.09s

Mean trip duration achieved
(with a minimal 1.31 vehicle
queue).



March 23-24: Finalised scenario design and statistical validation.

March 24: Successfully revised and resubmitted manuscript: Topography-Aware Deep Reinforcement Learning with Contextual Reward Engineering...

March 25: Prepared a concise transfer of manuscript results highlighting the adaptive traffic-management solution for U_CAN dissemination in Khmelnytskyi.

U_CAN Conference Preparation



Session Coordination and Stakeholder Engagement

Session I: Low-Emission Mobility	Session II: Municipal Measures	Session III: Citizen Engagement
5 March	6 March	9 March
Contacted invited European and academic experts to coordinate provisional titles and define the precise scope of their planned contributions.	Continued targeted email communication with Ukrainian municipal representatives and transport practitioners to confirm interest and participation.	Prepared and circulated additional invitations to academia, local government, and civil society, directly ahead of the U_CAN WP4 Monthly Meeting.

Programme Consolidation and Follow-up Actions



Project Reporting and Final Agenda Updates



Ukraine towards Carbon Neutrality



Conference Agenda

U_CAN

Ukraine towards Carbon Neutrality

National Conference on CO₂ Emissions Reduction in

Ukrainian Cities:

Pathways to Climate Neutrality

Khmelnytskyi | HYBRID | April 24, 2026 (EEST)

Month 24 of the Horizon Europe Project U_CAN

(GA No. 101148374)

Time	Session / Topic	Speaker	
09:30–10:00	Registration	KhNU Organizing Team	
10:00–10:30	Opening Session—Ukraine on the Path to Carbon Neutrality	10:00–10:10	Serhii Matiukh , Rector (KhNU)
		10:10–10:20	Kateryna Skyba , Vice-Rector (KhNU)
		10:20–10:25	Oleg Sinyuk , Vice-Rector (KhNU)
		10:25–10:30	Vavryshchuk Mykola , Deputy Mayor of Khmelnytskyi (KhCC)
10:30–12:00	Keynote Address—Reducing CO ₂ Emissions through Sustainable Urban Mobility	10:30–11:00	Swati Kulashri , Online, Project U_CAN Coordinator (TUD)
		11:00–11:30	Mobilities for EU , Online, (TBA)
		11:30–12:00	Jonas Fahlbusch , Online, Deputy Managing Director (Mobility2Grid)
12:00–13:00	Session I—European Solutions for Low-Emission Mobility	12:00–12:20	Maciej Skulimowski , Online, Research Assistant (WUST)
		12:20–12:40	Helle Ørsted Nielsen , Online, Senior Researcher (AU)
		12:40–13:00	Yurii Polianskyi , Project Manager (City Institute Lviv)
13:00–14:00	Lunch break		



Conclusions and Ongoing Future Work

- **Campus Venue Secured:** The designated conference venue on the Khmelnytskyi National University campus is nearly ready to host the hybrid event.
- **On-Site Logistics:** Final arrangements for essential on-site services, including catering, video broadcasting, sound systems, etc. are currently being completed.
- **Speaker Templates:** Official presentation templates are being finalised and will be provided to all speakers shortly.
- **Abstract Finalisation:** We will contact all confirmed participants soon to finalise their exact presentation topics and collect their abstracts.
- **Final Countdown:** Over the next two weeks, completing these operational steps will ensure a flawless execution of the U_CAN conference on 24 April 2026.





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Thank you!



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